INCH-POUND

MIL-DTL-25531F
31 July 2013
SUPERSEDING
MIL-DTL-25531E
05 June 1996

DETAIL SPECIFICATION

COUPLING HALF, QUICK DISCONNECT, PNEUMATIC STARTER, EXTERNAL, LOW PRESSURE

INACTIVE FOR NEW DESIGN

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AMSC N/A FSC 4730

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This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for quick-disconnect external low pressure half couplings for a pneumatic starter, with primary and secondary methods of release and, automatic pressure lock with self-closing valves.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

RR-C-271	-	Chains and Attachments, Carbon and Alloy Steel

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-DTL-22706	-	Duct and Scuff Cover, Pneumatic, Flexible
MIL-DTL-32258	-	Nut, Self-Locking (Ring Type Non-Metallic Insert), Heavy Hex, Controlled Root Radius, Nickel-Copper Alloy
MS17833	-	Connector, Pneumatic Starter Duct, Flanged, Male
MS33740	-	Nipple, Pneumatic Starting, 3 Inch ID

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-889	-	Dissimilar Metals
MIL-STD-31000	-	Technical Data Packages

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(Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION of AMERICA, INC. (AIA)

NASM20995	-	Wire, Safety or Lock
NASM33540	-	Safety Wiring and Cotter Pinning
NASM90725	-	Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A

(NA/NAS documents may be obtained online at www.aia-aerospace.org/ or from Aerospace Industries Association 1000 Wilson Boulevard, Suite 1700 Arlington, VA 22209-3928.)

SAE INTERNATIONAL

SAE-AS9581	-	Washer, Key-Cres, AMS 5510. 90 DEG. Locking

(SAE documents may be obtained online at http://www.sae.org/ or from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 USA.)

2.4 Order of Precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The coupling furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor’s drawings.

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3.2.1 Standard parts. With the exception specified in 3.2.a, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicide agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicide treatment will not be necessary.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions which may be encountered in storage or normal service.

3.3.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design. The coupling shall conform dimensionally to Figure 1 and shall be in accordance with the following requirements. One end of the coupling shall mate with the MS33740 nipple and the other end shall mate with the flange of hose and fitting MS17833. No part of the coupling shall protrude into the nipple bore. Three holes shall be provided as stated in paragraph 3.5.2 for possible attachment of handling ring.

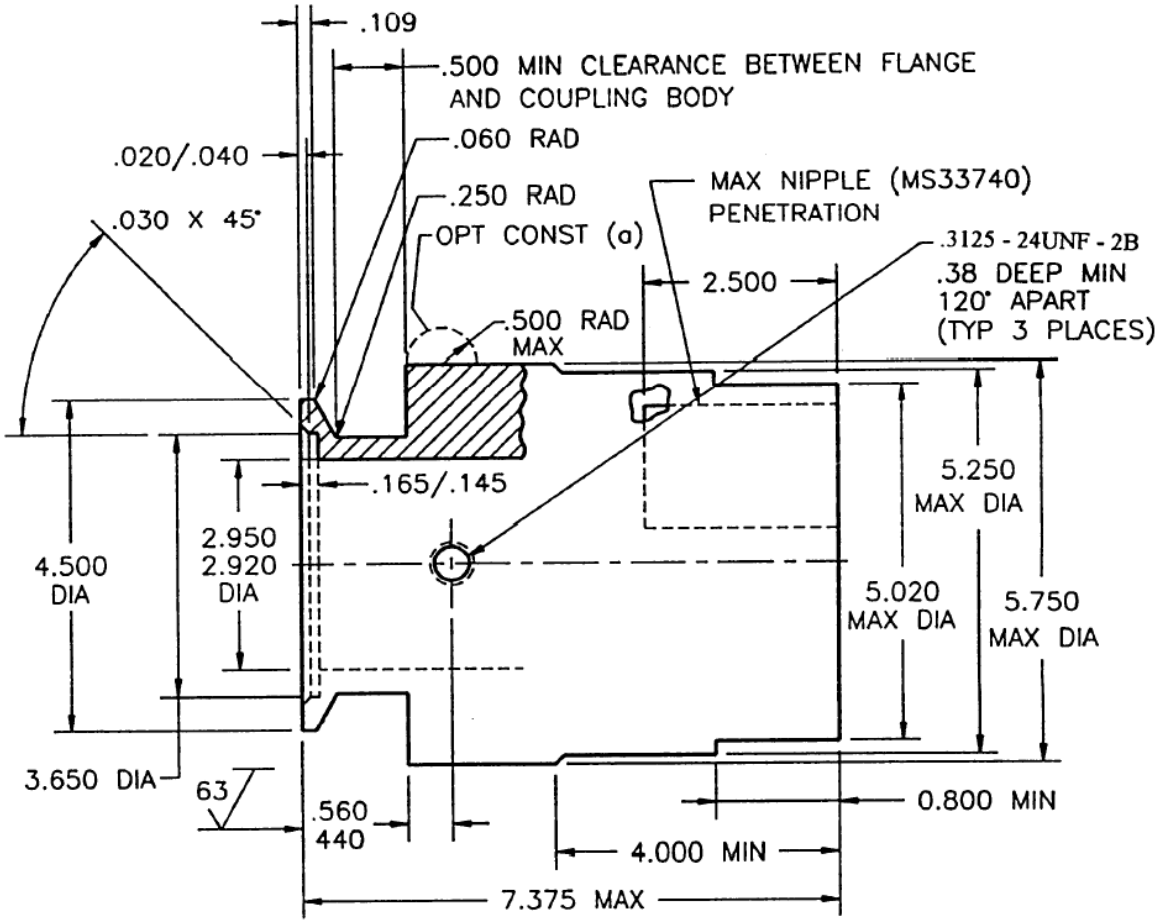
3.4.1 Reliability. The coupling shall be designed and constructed as specified herein in a manner that will insure the highest degree of operational reliability under all service conditions.

3.4.2 Instructions. Brief appropriate operating instructions such as "PUSH COLLAR FORWARD TO LOCK," or a directional arrow shall be provided as a permanent part of the coupling, preferably a part of the outer coupling casting.

3.5 Construction. The coupling shall be so constructed that it will withstand the strains, jars, vibrations, and other conditions incident to installation and service usage.

3.5.1 Maintainability. The coupling shall have a minimum number of parts consistent with reliability. Its design shall permit easy assembly, disassembly, location of trouble sources, and maintenance with tools normally available commercially, by service maintenance personnel with a minimum of training.

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(a) Protuberance shall not extend more than 2 inches along circumference at outside radius of body of coupling.

Flange fitting end piece to mate with flange of MS17833.

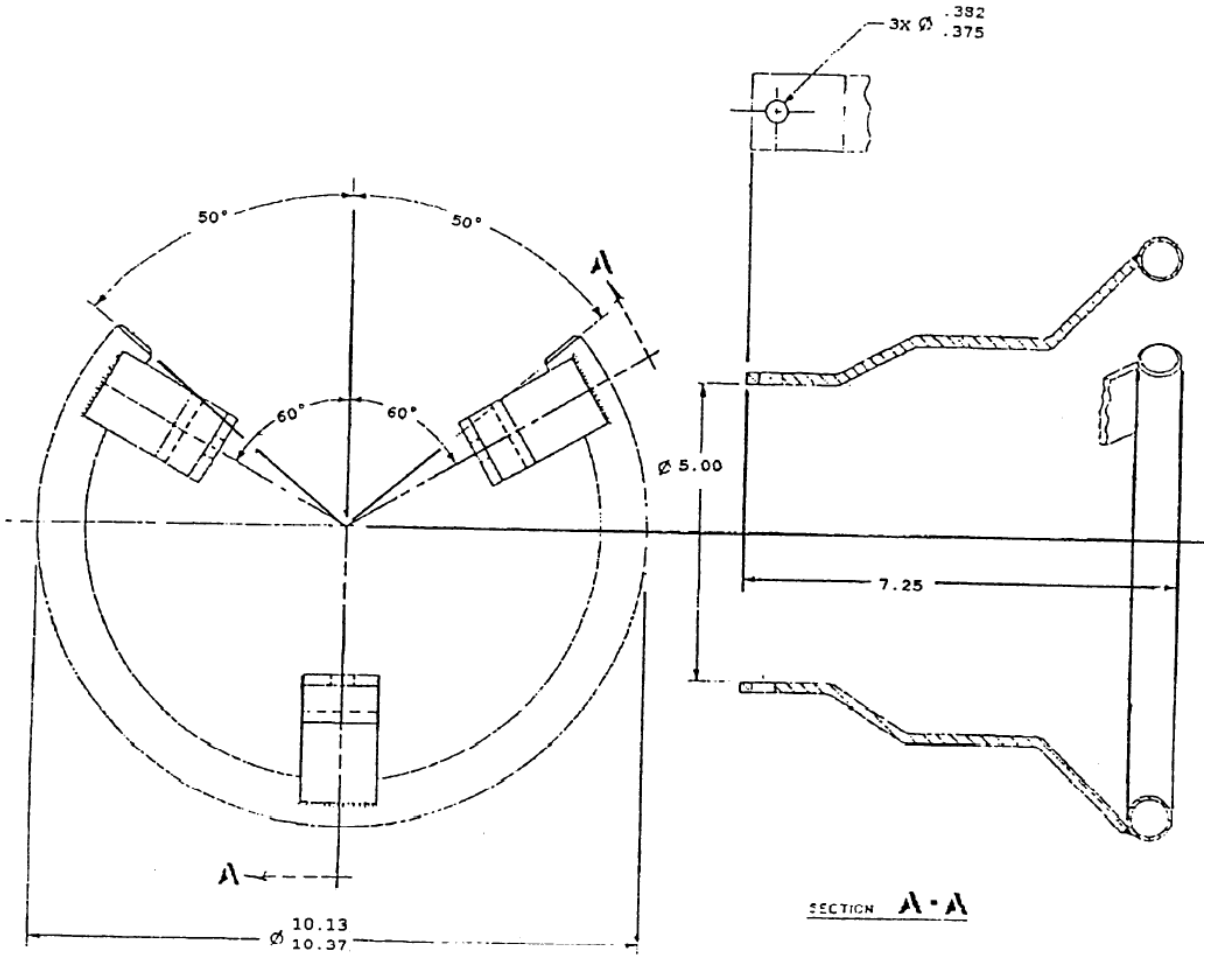
Dimensions in inches, unless otherwise specified, tolerances ±.010 inch.

FIGURE 1. Coupling half, quick disconnect, pneumatic starter, external, low pressure.

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3.5.2 Handling ring attachment holes. The coupling housing (collar) shall contain three holes. The holes shall be spaced 120 degrees apart, circumferentially, with one of the holes located 180 degrees circumferentially from the point of the handling ring attachment. Each hole shall be located axially .460/.540 inches from end of the coupling housing and shall be tapped for .3125 - 24UNF - 2b threads of MIL-S-7742 to a minimum depth of 0.380 inch (See Figure 1).

3.5.3 Handling ring. Ring is to be dimensional such that it will not interfere with aircraft structure or equipment attached to the aircraft. Configuration shall be approved by the specification preparing activity. Ring is currently available under National Stock Number (NSN) 5340-01-323-5523 (Figure 2). Corrosion resistant steel attachment bolts, per NASM90725 , and tab washers will be used to attach the handle to the coupling housing. Tab washers shall be SAE-AS9851 or equivalent.



NSN 5340-01-323-5523

FIGURE 2. Handling ring.

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3.6 Locking of parts. All threaded parts shall be locked by safety wiring, by self-locking nuts conforming to MIL-DTL-32258, cotter pin, or other approved methods. Safety wire shall be installed in accordance with NASM33540 and shall conform to NASM20995. The use of lock washers, set screws or staking is not permitted unless specifically approved by the using service. However, staking shall not be used where failure of the staking will allow any loose part in the flow stream.

3.6.1 Coupling to fitting connection. Provision shall be made for positive locking of the hose fitting to the coupling. The lock shall be of a reusable type.

3.7 Performance. The coupling shall function equally well with air flowing in either direction and shall satisfy the applicable requirements when tested in accordance with 4.6.

3.8 Attachment. The force required to engage the coupling to the nipple shall not exceed 25 pounds. Attachment shall be made by inserting the coupling onto the nipple by virtue of the handling ring (National Stock Number (NSN) 5340-01-323-5523) attached to the coupling collar. The coupling shall be so constructed that in the unattached position it is ready for instant connection to the aircraft nipple. It shall not be necessary to first retract the sleeve for connection. The coupling shall also be so constructed that it will not become partially locked on the nipple, allowing it to come off during the operation. Pulling on the hose normally connected to the coupling shall not result in disengagement of the coupling.

3.9 Release. The coupling shall contain two methods of release from the aircraft nipple, a primary and secondary release.

3.9.1 Primary. Primary release of the coupling shall be in pulling back on the handling ring and not requiring a force of more than 30 pounds to release the coupling from the aircraft nipple. This release shall leave the coupling in position for the next attachment.

3.9.2 Secondary. In the event the handling ring is not in use, disconnect can be made by simply grasping the coupling sleeve and pulling it in the reverse direction. If for some reason a lanyard is required such shall be constructed of a chain complying with or equivalent to RR-C-271, Type 1, Grade C, Class 5, Style 2, zinc coated, .15625 (Number 2) size with a T-type handle attached to the free end. End connecting links shall be of equivalent welded and zinc coated steel wire. The handle and 2 inch chain adjacent to the handle shall be covered with a heat resistant plastic or rubber covering.

3.9.3 Safety device. An automatic pressure lock and a self-closing automatic mechanical safety valve shall be provided for safety of operation. The pressure lock shall immobilize the primary and secondary release when the coupling is in the engaged position and under pressure. The pressure lock shall be so located that it is always accessible without disassembly of the coupling for manual deactivation while the coupling is connected to the nipple. The automatic pressure lock shall prevent disconnect of the coupling when the pressure is above 10 psi. It shall only be necessary for the valve to sufficiently stop the air flow to prevent danger to operating personnel. The device incorporated shall not be adversely affected by gravitational forces.

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3.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-31000 shall govern changes in the manufacturer's part numbers.

3.11 Dimensions. The dimensions of the coupling shall be as shown in Figure 1.

3.12 Weight. The weight of the coupling shall be the minimum consistent with required performance, but shall not exceed 10 pounds.

3.13 Finishes and protective coatings.

3.13.1 Finishes. The coupling shall have no painted surfaces.

3.13.2 Protective coatings. The coupling shall be adequately protected against corrosion by use of material or protective coatings acceptable to the procuring activity.

3.13.2.1 Anodizing. All aluminum-alloy parts shall be anodized in accordance with MIL-A-8625.

3.14 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.15 Workmanship. The coupling shall be uniform in quality, clean, sound, smooth, and free from injurious defects.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.2).

4.2 Classification of tests. The inspection and testing of the couplings shall be classified as follows:

a. Qualification tests (see 4.3)
b. Quality conformance test see 4.4).

4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6 and conducted in the order listed.

4.3.1 Sampling instructions. The qualification test samples submitted for Government tests shall consist of two couplings of each manufacturer's part number upon which qualification is desired. Samples submitted shall be accompanied by a report of the completed qualification test and shall

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be identified with the manufacturer's part number and any additional information required by the letter of authorization.

4.4 Quality conformance tests. Quality conformance test shall consist of individual tests and sampling tests.

4.4.1 Individual tests. Each coupling shall be subjected to examination of product (see 4.6.1).

4.4.2 Sampling tests.

4.4.2.1 Lot. A lot shall consist of couplings of the same part number manufactured under essentially the same conditions and submitted for inspection at the same time.

4.4.2.2 Sampling. One coupling shall be selected at random from each lot of 50 or fraction thereof and subjected to the following tests:

a. Engagement and release (see 4.6.2)

b. Pressure (see 4.6.3)

c. Preservation, packaging and marking (see Section 5).

4.4.3 Rejection and retest. When one or more couplings from a lot fail to meet the specification, acceptance of all couplings in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.4.3.1 Individual test may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of this specification.

4.4.3.2 Defects in couplings already accepted. The investigation of a test failure could indicate that defects may exist in couplings already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions.

4.5.1 Standard atmospheric conditions. Unless otherwise specified, tests shall be made at atmospheric pressure (approximately 29.92 inches of mercury) and at room temperature (approximately 77°F). When tests are made where atmospheric pressure or room temperature differ materially from these values, proper allowance shall be made for the difference from the specified condition.

4.5.2 Cycles. Unless otherwise specified, a total of 10 cycles shall be accomplished for all release tests.

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4.5.3 Lubrication. No lubrication or other servicing of the coupling or mating nipple shall release tests.

4.5.4 Test nipple. In all tests where a nipple is required, the nipple shall be in accordance with MS33740.

4.5.5 Ducting. Where ducting is attached to the coupling for test purposes, the hose end fitting shall conform to MS17833 and the ducting shall conform to MIL-D-22706.

4.6 Test methods.

4.6.1 Examination of product. The coupling shall be examined to determine compliance with respect to materials, workmanship, dimensions and markings.

4.6.2 Engagement and release. The coupling shall engage a nipple when a maximum force of 25 pounds or a maximum torque of 25 lb.-ft, as applicable, is exerted. The coupling shall release from the nipple by the secondary method when a force of 10 to 40 pounds is exerted. The coupling shall disconnect from the nipple when the lanyard release is pulled with a force of 10 to 40 pounds within a minimum cone angle of 30 degrees.

4.6.3 Pressure. The coupling, mounted on a test nipple, shall be pressure tested at the pressures listed below. Each pressure shall be held for 5 minutes. Leakage shall not exceed 1 pound per minute (pm). The coupling shall be capable of normal operation after completion of this test.

a. Operating pressure - 112 psia at 600°F

b. Proof pressure - 200 psia at 600°F.

4.6.4 Pressure drop. Pressure drop through the coupling, when attached to the nipple, shall not exceed 0.9 psi when 450°F air is supplied at 55 psi static pressure and 120 pm flow.

4.6.5 Temperature operation. The coupling shall be engaged and disengaged with a nipple at the following temperatures. Failure to successfully engage and disengage a total of five consecutive times for each temperature condition shall be cause for rejection.

a. Nipple and coupling at -65°F

b. Nipple at -65°F and the coupling at +600°F internal surface temperature

c. Nipple at 400°F and coupling at 600°F internal surface temperature.

4.6.6 Drop tests. The coupling, while attached to ducting, shall be engaged with a nipple mounted in a horizontal position 6 feet above a concrete surface as measured from the centerline of the nipple. The internal surface temperature of the coupling shall be raised to 600°F. The coupling shall be disconnected by the handling ring or by tripping the lanyard release and allowing it to fall on the concrete surface. This cycle shall be repeated 125 times with the

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coupling being rotated 72 degrees after every 25 drops and reheated to an internal surface temperature of 600°F after every 25 consecutive drops. This procedure shall be repeated for 125 disconnects when the coupling and nipple are mounted in the vertical position with the flange and nipple 6 feet above the surface. Upon completion of this test, the coupling shall be subjected to the test specified in 4.6.3.

4.6.7 Safety device.

4.6.7.1 Safety valve. The proper and reliable operation of the automatic self-closing valve shall be demonstrated in accordance with 4.5.2. The coupling attached to one end of a 10 to 30 foot length of ducting shall be placed on level concrete with the duct making a large radius angle of 90 degrees and the coupling end free to move. Air at 112 psia shall be applied to the duct. After initial straightening of the duct, there shall be no perceptible movement of the coupling end of the duct. The proper operation of the safety valve shall be demonstrated by the following procedure. With the pressure lock deactivated, the coupling attached to the nipple, and with air flowing under a pressure of 80 psia and stabilized at 600°F, handling ring actuation of the coupling shall cause the coupling to disconnect and instantaneously close the safety valve. As a safety precaution, the coupling end of the house shall be suitably restrained.

4.6.7.2 Pressure lock. The coupling, engaged with a nipple, shall be subjected to 112 psia. A coupling release shall be attempted at this pressure. The pressure shall then be reduced in increments of 10 psi and at each increment the handling ring shall be pulled three times with sufficient force to provide a component parallel to the coupling axis of 25 to 35 pounds. Disconnection of the coupling when a pressure above 10 psi is applied or failure of the pressure lock to operate satisfactorily as a result of this test shall be cause for rejection.

4.6.8 Endurance. The coupling shall satisfactorily complete the following endurance test, after which it shall be subjected to the test specified in 4.6.3. Only one nipple shall be used throughout this test. Any galling or distortion of the nipple that would be detrimental to proper engagement or release of the coupling shall be cause for rejection.

a. Cyclical: The coupling shall be engaged and disengaged to the nipple a total of 1,000 times. The engagement and disengagement shall be accomplished after the coupling has been subjected for 1 minute to an air flow of 100 to 188 pm at a total air pressure and temperature of 80 psia and 500°F, respectively. The cycle shall be conducted at intervals of approximately 4 minutes.

b. Continuous: The coupling shall be subjected to 50 cycles, each cycle being 2 hours duration. During 1 hour of the cycle, air shall be flowed through the coupling at the rate of 100 to 188 pm at a total air pressure and temperature of 80 psia and 600°F, respectively. During the second hour there shall be no air flow. The coupling shall be released from the nipple following the 1-hour flow period and shall be reengaged following the 1-hour no-flow period.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The couplings covered by this specification are intended for use with an aircraft mounted nipple for the purpose of connecting the aircraft with a ground service low pressure air supply for operation of an air turbine starter, air turbine motor or other pneumatically powered equipment. Use is limited to applications within the requirements of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-25531 whether or not such products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Oklahoma City Air Logistics Center/ENSDAA (AF-71), 3001 Staff Drive, Suite 1AB81A, Tinker AFB, OK 73145 or email tinker.dsp@tinker.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Requested data. When requested, the manufacturer will submit engineering drawings and inspection reports.

6.3.2 Attachment holes. The addition of attachment holes for MS14271 handling ring to a previously qualified coupling half will not be required to re-qualify.

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6.4 Subject term (key word) listing.

Air flow
Endurance
Handling ring
Nipple
Pressure lock
Primary release

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - AS
Air Force - 71

Preparing activity:
Air Force - 71

(Project 4730-2013-092)

Reviewer activities:
Air Force - 99
DLA - GS

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